

Software Requirements and Design Document

For

Code Worlds

Version 1.0

Authors:

Liam Sharkey

1. Overview (5 points)

Give a general overview of the system in 1-2 paragraphs (similar to the one in the project proposal).

Code Worlds is going to be an educational 2d RPG game, inspired by early Pokémon and online education games. Players will control a player character to explore the world, complete quests, including a main storyline, capture CodeMals, and engage in CodeMal battles. Throughout the game, players will have to complete coding minigames to move forwards. The defining feature of Code Worlds will be the coding minigame which may be involved at various points within the game, sometimes within battles or captures, and sometimes during exploration.

2. Functional Requirements (10 points)

*List the **functional requirements** in sentences identified by numbers and for each requirement state if it is of high, medium, or low priority. Each functional requirement is something that the system shall do. Include all the details required such that there can be no misinterpretations of the requirements when read. Be very specific about what the system needs to do (not how, just what). You may provide a brief design rationale for any requirement which you feel requires explanation for how and/or why the requirement was derived.*

- 1) Player movement and environment – High priority. The player will control a character and move them throughout the world. This feature is comparatively easy and has expansive precedent, but is nonetheless extremely necessary. This will be the core of how the player experiences the world and all other features.
- 2) Unique expansive world – Medium priority. The players need a world that feels engaging to explore and play in. The goal is to have unique assets, areas, and activities to engage with throughout a large world.
- 3) Quests and NPC's - High priority. There is to be an overarching main questline for the player to complete to 'finish' the game, along with side quests to give variety of activities. The main quest will have the player go to every corner of the world, fight various battles, and hopefully come to a better understanding of how to code. The side quests will be shorter and may only contain a couple fights, exploratory moments, and interactions. NPC's (non player characters) will be necessary to facilitate quests, as the most interesting part of any story is the characters. NPC's will have dialogue, move to match certain questlines, and provide rewards for players depending on progress. Fleshing out characters will be a relatively low priority compared to the other two.
- 4) CodeMals – High priority. The mechanics of CodeMals is the beating heart of the game. The battles, capture sequences, and interaction with the coding minigames will be what players remember most and will take the most time and effort. Players will collect, battle, and grow throughout the game with their CodeMals. The mechanics of these systems will be fleshed out over time as we determine what is most fun to interact with, but for now battles and captures will mostly resemble those from Pokémon, with coding challenges for an interesting twist.
- 5) Coding minigames – High priority. Players will be required to complete coding questions throughout the game in order to progress and succeed. Coding questions connected to questlines may be custom made and more difficult, while those encountered randomly may be procedurally generated through a custom system. The degree to which and where these minigames appear will be changed and experimented with throughout development, but will be a core mechanic as forward as the CodeMals themselves.
- 6) UI – Low priority. A basic UI has to exist for basic functionality, but a well designed and good looking UI is a low priority. Things such as a main menu, pause menu, clear buttons and interactables within the world, and other necessary elements will be implemented early on, and hopefully polished in the late stages of the project.
- 7) CodeMal Design – Low priority. Custom assets and designs of the CodeMals is considered a polish stage priority and not core the gameplay. Placeholder assets will be used for a majority of development, while custom art will be created later on. (This

category does not include design of their mechanics such as attacks or practical uses, only art direction)

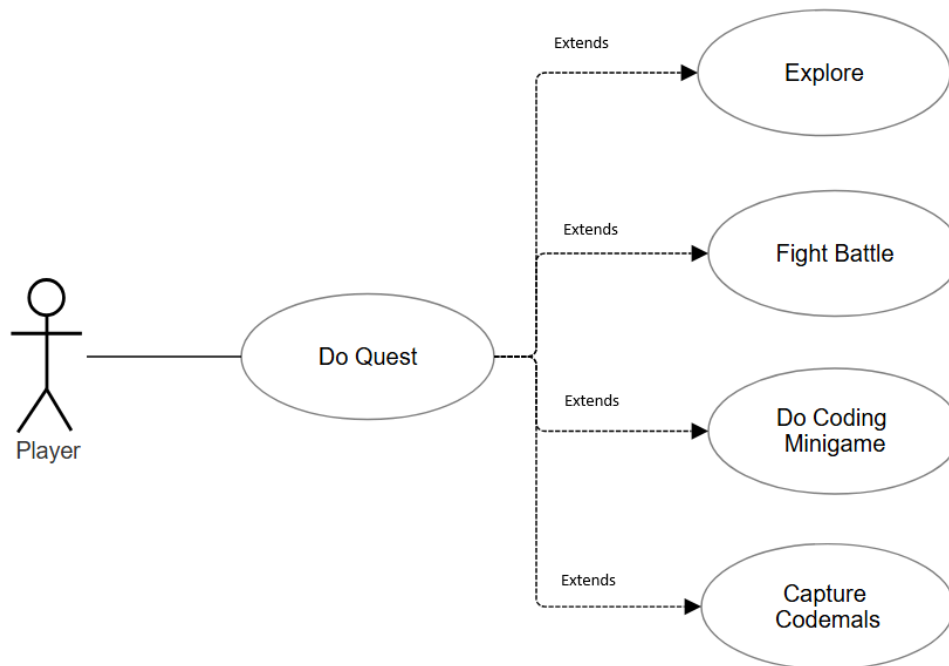
- 8) Save Game Functionality – Players will be able to save where they are in the game to their local machine, and have multiple saves at the same time to go back and forth between.

3. Non-functional Requirements (10 points)

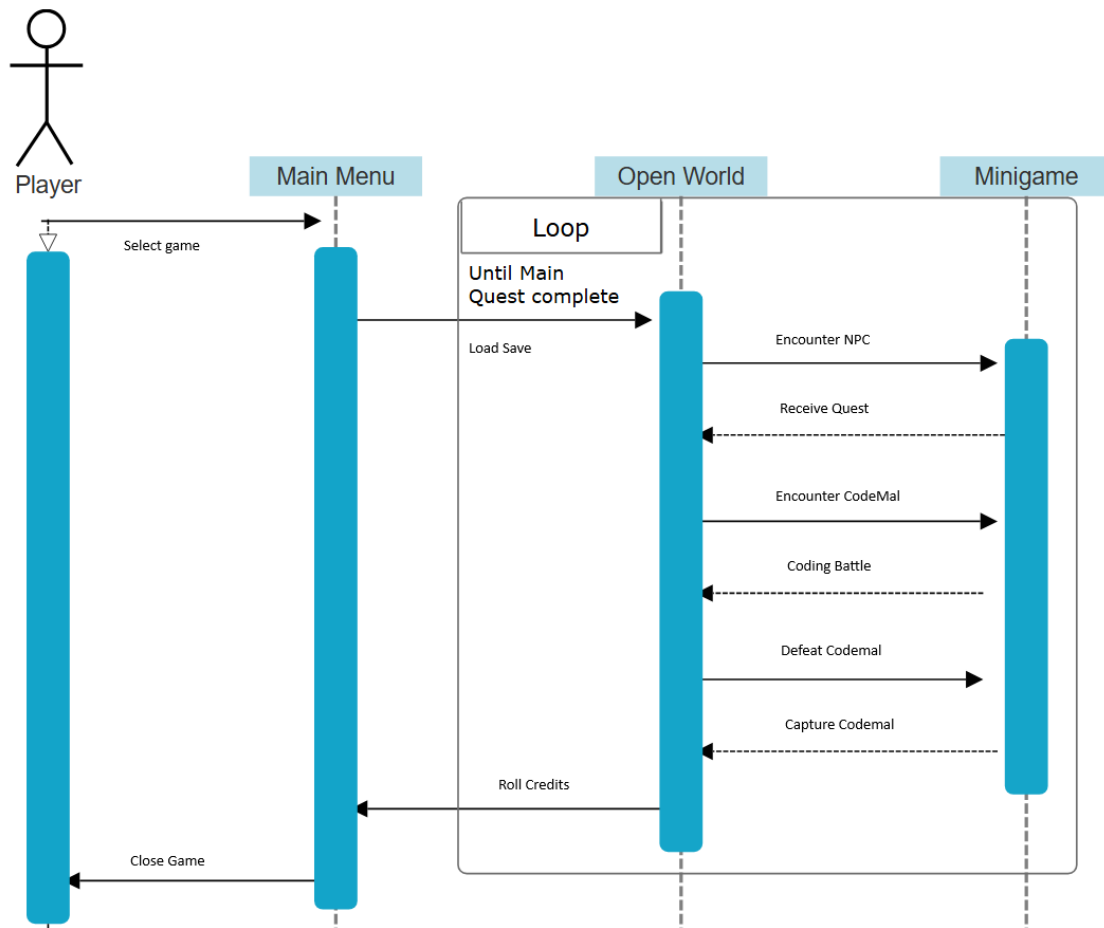
List the **non-functional requirements** of the system (any requirement referring to a property of the system, such as security, safety, software quality, performance, reliability, etc.) You may provide a brief rationale for any requirement which you feel requires explanation as to how and/or why the requirement was derived.

- 1) Easy to run – This is meant to be a game for anyone to learn how to code and should therefore run on any computer that someone might code on. The graphics will be basic, physics will be mostly nonexistent, and any feature that would jeopardize the ability for someone to learn to code will not be implemented.
- 2) Entirely local – There will be no need to connect to the internet beyond the download of the game. This will function as a classic single player game, so that someone may learn to code anywhere.

4. Use Case Diagram (10 points)



5. Class Diagram and/or Sequence Diagrams (15 points)



6. Operating Environment (5 points)

This will be a desktop game. Godot supports mac, windows, and linux, so it is intended to work on all 3, though only on modern version of said systems. The game will run locally and save locally, it will not interact with any external systems other than the operating system.

7. Assumptions and Dependencies (5 points)

This project assumes continued support by Godot, along with the ability for Godot to export games for the 3 listed operating systems. We also assume the existence of a free asset library for development, though we don't plan to keep these assets for the entirety of the project. We also assume that Nintendo doesn't sue us like they do everyone else.